

IT Concepts

Wednesday, January 17, 2024

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Hardware

- The physical components - such as System Unit, Display unit, keyboard

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PC/Desktop

- Personal computer - general-purpose computer- price, capabilities suitable for individual user

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Two types of PC

- IBM compatible PC & PC from Apple
- Both have different OS, hardware and Applications. Application for IBM PC does not run on Apple PC
- Home users, large and small office users

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Laptop

- Portable computer
- Run on batteries
- Do not use Monitor, has special screen as display unit

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Notebook

- Small version of laptop
- Typical users-Business users, people on the move, educational users.

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PDA

- Very compact
- Uses special pen for input
- Has some utility application - clock, calendar, calculator
- Some has internet connectivity
- PDAs can also be used for accessing the Internet, sending and receiving e-mails, video recording, making and writing on spreadsheets, scanning bar codes, playing computer games, recording survey responses, and Global Positioning System (GPS).
- Typical users : Business users

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Mobile phone

- used for mobile communication.
- As well a speech they may be used for text messaging, accessing web, e-mail
- Smart mobile phone has capacities like computer

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Smartphones

- A smartphone is a mobile phone (cell phone) offering advanced computer like features.
- Features vary one manufacturer to another
- Other features can include a built-in camera, the ability to play music, display photos and video clips, media software for playing music

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Media players

- Media players allow you to store digital music and video.

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Parts of a computer

- Central processing unit (CPU)
- Types of memory
- The hard disk
- Input and output devices

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CPU

- Performs all process
- Parts:
 - Control Unit : responsible for controlling the sequencing and timing of the other elements in the CPU
 - Arithmetic Logic Unit: performs the mathematical calculations using data stored within the CPU registers
 - Register: memory storage areas within the CPU that hold the data used by ALU
 - Bus: transports data between the memory and registers

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Memory (RAM)

- Primary memory
- Any application we run must be loaded in the RAM. Even OS is loaded in the RAM

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ROM-BIOS (Read Only Memory - Basic Input Output System)

- A special chip held on your computer's system (mother) board.
- Responsible copying your operating system into RAM

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Disks

- any data/file that you create is stored in disks

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Hard (Fixed) Disk

- Operating system, applications we install are stored on hard disks
- All the files we create are stored here

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Universal Serial Bus (USB) Port

- Used to connect flash drives, camera, printers, scanners etc.
- Most devices we connect through USB are plug and play devices
- Allows data transfer and supply power to devices

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Serial Port

- Used to connect devices like modem
- Labelled as COM1 & COM2

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Parallel Port

- Used to connect devices like printers
- Labelled as LPT1 & LPT2

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Network Port

- Used to plug a 'network cable' which allows communicate with other computers in local network or to access the Internet.

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Factors those affect computer performance

- CPU Speed - higher the speed the faster the computer will work
- RAM Size - all its working files are kept RAM and hence larger RAM allows running faster
- Graphics card processor and memory - To display what user wants, computer uses its graphics card.
If the graphics card memory is low the graphics card shares the RAM and user is not getting the RAM at its full performance
- Number of applications running - Windows is a multitasking system, it can run multiple programs at a time.
The more programs which are running at the same time, the slower each one will run

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Measurements

- 1 bit - it can store 0 or 1
- 1 byte = 8 bit
- 1 Kilobyte = 1024 bytes (1024*8 bits)
- 1 Megabyte = 1024*1024 bytes (1024*1024*8 bits)

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Computer memory

- Memory is used to store data
- Data in RAM temporary
- Data in hard disk is permanent

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RAM

- Random Access Memory (RAM) is the main 'working' or 'primary' or 'main' memory used by the computer
- Any application we run is loaded in the RAM first.
- Data in RAM is volatile

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ROM

- Read Only Memory (ROM)
- a special type of memory chip which holds software
- Data can be read from ROM but cannot write data

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ROM-BIOS

- The 'Read Only Memory-Basic Input Output System' a chip located on system (mother) board,
- Performs a self-diagnostic to check all the parts are working when we start the computer
- It also loads OS into RAM

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Video (graphics) memory

- The picture which you see on screen processed in video card.
- The on-screen pictures are held in video card memory

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Internal hard disks

Every computer has an internal hard disk.

Operating system, Applications we install are stored

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External hard disks

- Secondary storage that we plug into commonly via USB port
- Mostly used for backup photos, movies or any kind of data files
- CD (Compact disk)

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Optical media

- CDs are read using CD/DVD drive
- Used for supplying movies, music and games
- Size 650 MB

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DVDs

- Short for "Digital Versatile Disk".
- Also optical media
- CDs are read using DVD drive
- Size is much larger than CDs approximately 4GB
- Used for supplying movies, music, games, application software

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Recordable CD (CD-R)

- Same as CD but allows us to record data, music or video
- To write data we need CD/DVD writer

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USB flash drives (memory sticks)

- Portable drive plug into computer via USB port
- Mainly used for data transfer
- Size varies from 4GB to 128GB or more

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Memory cards

- Used in devices such as digital cameras, telephones, music players, video game consoles, GPS system and similar devices to store data
- Different types of memory cards with different storage capacities are available

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Network drive

- A storage location on another computer in same network or network server
- In an office, it provides central location for storing data
- It can used central location for backup data

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Input devices

- Input devices allow you to input information to the computer
- Different of input devices, each has different method to input

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Keyboard

- The keyboard allows you to type information into the computer
- Desktop computer uses external keyboard connected via cable
- Laptops has built-in keyboard

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Mouse

- Allows interact with computer via click on button .
- use the mouse to select drop down menus, to point and click on items, to select items and to drag and drop items from one place to another

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Scanner

- Scan printed material and convert it into a file format that used within the PC.

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OCR (Optical Character Recognition)

- Special programs convert printed text into editable text

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Tracker balls

- A tracker ball is an alternative to the traditional mouse
- Mostly used by graphics designers for better precisions

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Touch pads

- A touch pad is a input device responds to pressure.
- Mainly input are taken by touching, swiping, pinching etc

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- ? Joysticks
 - Mainly used by gamers
 - It has more button than a mouse
 - It button has different action
 - Responds to movement in 3 axis direction

Web cams

- Allows two-way communication
- Supports sound and video communication
- Used for video calling, online meeting, online class etc

? Digital cameras

- Used in the same way a traditional camera
- Do not require films instead stored in memory
- Pictures can be enhanced using graphic programs

? Microphones

- Used for voice input

? Flat screen computer screens

- Requires less space and use less energy than tradition monitor
- Use LCD for display

? Projection devices

- Used for presentation
- Quality and price depend on the resolution (go for a minimum of XGA) and the brightness of the lamp (the brighter the better)

? Speech synthesizers

- Coverts voice into text
- Most voice recognition programs uses Speech synthesizers

? Printers

- Prints documents, images

? Types of printers

- Inkjet- prints by spraying tiny ink jet onto the paper.
- Laser- used drum and press technology to print , quality and speed is better than inkjet printers
- Dot-matrix-uses pin and ribbon. Very cheap printer and noisy.

? Input and output

- Some devices are both input and output
- For example - modem, touch screen

? What is software?

- Software is the collection of instructions which makes the computer work

? Operating system

- Every computer must have an operating system to work
- It manages all hardware and application
- Example: Windows 10, Windows 11, Linux, Mac OS

? Application Software

- Each application is for specific purpose
- Users install application software according to their needs

? Word processing

- A word processing program (such as Microsoft Word) allows you to produce letters, memos, etc., easily.
- It allows easily creating personalized letters to customers or subscribers individually using mail merge

? Spreadsheet

- A spreadsheet program (such as Microsoft Excel) allows you to work out a company's income, expenditure and then calculate the balance.
- It has formula, function for processing which updated automatically for value changes

? Database applications

- A database program (such as Microsoft Access) allows to create database, manage data (read, write, update) securely and efficiently.

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Presentation

- A presentation program (such as Microsoft PowerPoint) allows you to produce professional looking presentations which can be printed, display as slides on projector/computer display.

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Photo editing applications

- These programs allow you to edit digital photos.

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Operating Software VS Application Software

- Operating System is often known system software
- It manages hardware and software
- Every computer must have an operating to work
- Application software are installed on top of operating system
- An application software is for specific tasks.
- Users install application software based on their need

Options to improve accessibility

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Speech recognition

- Allows to send command using voice.
- This type of software is commonly used when phoning large organisations, typically banks to recognize your card number from voice and act other task.

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Narrator - Screen reader software

- Reads out content of the screen
- Designed for visually impaired users

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Screen magnifier software

- Magnifies the screen contents